# Fizika

9-sinf

## Mavzu: Masalalar yechish.

$$F = \sigma l \Rightarrow \sigma = \frac{F}{l}$$

$$W = \sigma S \quad pufakda: \quad W = 2\sigma S$$

$$h = \frac{2\sigma}{\rho_s g r}$$

# 20-mashq 1-masala

Suv kapillyarda 14 mm ga ko‘tarilgan bo‘lsa, uning diametri qanday bo‘lgan?

$$g = 10\ m/s^2,\ \ \rho_s = 10^3\ \frac{kg}{m^3},\ \ \sigma = 73 \cdot 10^{-3}\ N/m.$$

## Berilgan:

$h = 14\,mm = 14 \cdot 10^{-3} m$

$\rho_s = 10^3\ kg/m^3$

$g = 10\ m/s^2$

$\sigma = 73 \cdot 10^{-3}\ N/m$

**Topish kerak:** $d - ?$

## Formula:

$$h = \frac{2\sigma}{\rho_s g r}$$

$$r = \frac{2\sigma}{\rho_s g h} \qquad r = \frac{d}{2}$$

$$\frac{2\sigma}{\rho_s g h} = \frac{d}{2}$$

$$d = \frac{4\sigma}{\rho_s g h}$$

**Yechish:** $$\mathrm{d} = \frac{4 \cdot 73 \cdot 10^{-3}\ \frac{N}{m}}{10^3 \frac{kg}{m^3} \cdot 10 \frac{m}{s^2} \cdot 14 \cdot 10^{-3} m} \approx 2 \cdot 10^{-3} m \approx 2\ mm$$

**Javob:** $\mathrm{d} \approx 2\ mm$

# 20-mashq 4-masala

Teshigining diametri 2 mm bo‘lgan tomizg‘ichdan tomadigan tomchining massasini aniqlang.

$$g = 10\ m/s^2, \qquad \rho_s = 10^3\ \frac{kg}{m^3}, \qquad \sigma = 73 \cdot 10^{-3}\ N/m.$$

## Berilgan:

$d = 2\,mm = 2 \cdot 10^{-3}\,m$

$g = 10\,m/s^2$

$\sigma = 73 \cdot 10^{-3}\,N/m$

$\pi = 3{,}14$

**Topish kerak:** $m - ?$

## Formula:

$$F_{og'} = F_{s.t.}$$

$$mg = \sigma l$$

$$m = \frac{\sigma l}{g}$$

$$l = \pi d$$

$$m = \frac{\sigma \pi d}{g}$$

**Yechish:** $$m = \frac{73 \cdot 10^{-3}\frac{N}{m} \cdot 3{,}14 \cdot 2 \cdot 10^{-3}\,m}{10\,m/s^2} \approx 46 \cdot 10^{-6}\,kg \approx 46\,mg$$

**Javob:** $m \approx 46\,mg$

# 20-mashq 6-masala

Zichligi $0{,}9\ g/cm^3$ bo‘lgan suyuqlik diametri 1,5 mm bo‘lgan kapillyar naydagi ko‘tarilish balandligi 10 mm bo‘lsa, shu suyuqlikning sirt taranglik koeffitsiyentini aniqlang.

## Berilgan:

$h = 14\,mm = 14 \cdot 10^{-3} m$

$\rho_s = 0{,}9\,\frac{g}{cm^3} = 0{,}9 \cdot 10^3\,\frac{kg}{m^3}$

$g = 10\,m/s^2$

$d = 1{,}5\,mm = 1{,}5 \cdot 10^{-3}\,m$

**Topish kerak:** $\sigma - ?$

## Formula:

$$h = \frac{2\sigma}{\rho_s g r}$$

$$\sigma = \frac{\rho_s g r h}{2}$$

$$r = \frac{d}{2}$$

$$\sigma = \frac{\rho_s g d h}{4}$$

**Yechish:**

$$\sigma = \frac{0{,}9 \cdot 10^3 \frac{kg}{m^3} \cdot 10 \frac{m}{s^2} \cdot 1{,}5 \cdot 10^{-3}\,m \cdot 14 \cdot 10^{-3} m}{4} =$$

$$= 0{,}04725\,N/m$$

**Javob:** $\boldsymbol{\sigma = 0{,}04725\,N/m}$

# 20-mashq 8-masala

Yer sirtidagi kapillyarda suv 15 mm ga ko‘tariladi. Agar Oyda erkin tushish tezlanishi Yerdagidan 6 marta kichik ekanligi ma’lum bo‘lsa, Oyda shu kapillyarda suv qanday balandlikka ko‘tariladi?

## Berilgan:

$h_y = 15\ mm$

$\frac{g_y}{g_o} = 6$

**Topish kerak:** $h_o - ?$

## Formula:

$$h = \frac{2\sigma}{\rho_s g r}$$

$$h_y = \frac{2\sigma}{\rho_s g_y r} \qquad h_o = \frac{2\sigma}{\rho_s g_o r}$$

$$\frac{h_y}{h_o} = \frac{g_o}{g_y} \Rightarrow h_o = \frac{h_y g_y}{g_o} = h_y \cdot \frac{g_y}{g_o}$$

**Yechish:** $h_o = 15\ mm \cdot 6 = 90\ mm$

**Javob:** $\boldsymbol{h_o = 90\ mm}$

# 20-mashq 9-masala

Sovun puvakchasining sirt yuzasi $12\ cm^2$ ga ortganda sirt energiyasi qanchaga o‘zgaradi? $\sigma = 0{,}04\ \frac{N}{m}$.

**Berilgan:**

$\sigma = 0{,}04\ N/m$

$\Delta S = 12\ cm^2 = 0{,}0012\ m^2$

**Topish kerak:** $\Delta W - ?$

**Formula:**

$$\Delta W = 2\sigma\Delta S$$

**Yechish:** $\Delta W = 2 \cdot 0{,}04 \frac{N}{m} \cdot 0{,}0012\ m^2 = 0{,}000096\ \text{J} = 96\mu\text{J}$

**Javob:** $\Delta W = 96\ \mu J$

# Mustaqil bajarish uchun topshiriqlar

102-bet  2, 3, 5, 7-masalalarni yechish.